



Jan Hendrik Metzen

Research Scientist

I am a Research Scientist at [Prior Labs](#), developing the next generation of tabular foundation models. My work focuses on scalable pretraining, novel architectures, and scaling laws, contributing to TabPFN-3's expansion to datasets with up to one million training rows and substantially faster training and inference.

From 09/2024 to 01/2026, I was a Senior AI Researcher in [Aleph Alpha Research](#)'s Foundation Models team, where I focused on LLM pretraining and optimization. I contributed to tokenizer-free LLM architectures for efficient pretraining, domain adaptation, and inference, as well as optimization methods for efficient training of large models.

I was Senior Expert at Bosch Center for Artificial Intelligence (BCAI) until 08/2024. As a Senior Expert in Deep Learning at [Bosch Center for Artificial Intelligence \(BCAI\)](#), my primary research focused on making AI (specifically computer-vision based perception) robust, reliable, and safe. This included [identifying vulnerabilities of Transformer-based neural network against adversarial patch/token attacks](#) as well as finding [systematic errors of image classifiers on rare subgroups](#) and [systematic errors of object detectors](#). We also developed architectures that are certifiably robust against patch attacks for [image classifiers](#) as well as for [semantic segmentation](#). Furthermore, we proposed methods for adversarially training neural networks to become robust against [universal perturbations](#) and [universal adversarial patches](#). Moreover, we provided methods for [test-time adaptation of neural networks to improve robustness to domain shifts](#) and studied the role of [shape-biased representations on robustness to common image corruptions](#).

A different strand of my research is automating machine learning (AutoML), specifically Neural Architecture Search. The latter research field is motivated by the vast design space of neural networks and the diversity of inference hardware. Manually tailoring a neural architecture for every type of hardware is cumbersome and not scalable - hardware-aware neural architecture search can vastly improve design efficiency and thus reduce cost of AI development. See our survey on [Neural Architecture Search](#) and a more recent survey on [Neural Architecture Search for Dense Prediction Tasks in Computer Vision](#). We have also developed [AutoCLIP](#), a method auto-tuning zero-shot classifiers for vision-language models that improves zero-shot performance across a broad range of domains.

I also love contributing to machine learning libraries, both open source and proprietary. I am a (recently mostly inactive) [core contributor of scikit-](#)

[learn](#), where I contributed tools for [probability calibration of classifiers](#) and for [kernel ridge regression](#). Moreover, I have written a complete redesign of the [Gaussian process module](#) for scikit-learn. At BCAI, I was involved as a core developer of frameworks for deep learning training pipelines, neural architecture search, and robustness evaluation.

I am an [ELLIS](#) member and an ELIZA Industrial Fellow. I serve as an Area Chair for [NeurIPS 2026](#) and regularly review for scientific conferences and journals such as ICLR, ICML, NeurIPS, and TMLR. I was senior area chair of the [AutoML 2022 conference](#) and co-organizer of the workshops [NAS@ICLR 2020](#) and [NAS@ICLR 2021](#). I have been recognized by ICLR as [Highlighted Reviewer in 2022](#) and [Outstanding Reviewer in 2021](#).

See my [personal website](#) for more details.

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🐙 [jmetzen](#) (github)

👤 [w047VfEAAAAJ](#) (Google Scholar)

Work **Prior Labs**

Research Scientist

Feb 2026 – Present

Research Scientist developing the next generation of tabular foundation models. My work focuses on scalable pretraining, novel architectures, and scaling laws, contributing to TabPFN-3's expansion to datasets with up to one million training rows and substantially faster training and inference.

- Pretraining and scaling laws for tabular foundation models
- Novel architectures for scalable and efficient tabular prediction

Aleph Alpha Research

Senior AI Researcher

Sep 2024 – Jan 2026

Senior AI Researcher in Aleph Alpha's Foundation Models team, where I focused on LLM pretraining and optimization. I contributed to tokenizer-free LLM architectures for efficient pretraining, domain adaptation, and inference, as well as optimization methods for efficient training of large

models.

- Efficient LLM pretraining
- Tokenizer-free LLM Architectures

Bosch Center for Artificial Intelligence (BCAI), Robert Bosch GmbH

● Senior Expert 'Robust Scalable Perception'

Jan 2020 – Aug 2024

Making Deep Learning Perception Safe by Model Auditing and Robust Training and Architectures.

- Robustness of Deep Learning (Systematic Errors, Domain Shift, Adversarial Examples)
- Neural Architecture Search
- Synthetic Data

● Research Scientist

Mar 2016 – Dec 2019

- Adversarial Examples
- Neural Architecture Search

Robotics Research Group, University of Bremen

Research Assistant

Nov 2009 – Feb 2016

- Reinforcement Learning
- Robot Learning
- Bayesian Optimization

Robotics Research Group, University of Bremen

Project Leader 'CASCADE'

Aug 2014 – Jan 2016

Leading the subteam in Bremen of the EU project CASCADE

- Surgical Robotics

Robotics Innovation Center, German Research Center for Artificial Intelligence (DFKI RIC, Bremen)

- **Team leader 'Sustained Learning'**

Jan 2013 – Dec 2015

Leading a team of approximately 10 researchers focusing on robotic learning.

- Robot Learning

- **Researcher**

Jul 2007 – Oct 2009

- Brain-Computer Interfaces
- Robot Learning

Robotics Research Group, University of Bremen

Research Assistant

Sep 2006 – Jun 2007

- Reinforcement Learning

Volunteer

NeurIPS 2026

Area Chair

Jan 2026 – Dec 2026

ELIZA

Industrial Fellow

Jan 2026 – Present

scikit-learn

Core Contributor

Jan 2016 – Dec 2023

AutoML Conference 2022

Senior Area Chair

Jan 2022 – Dec 2022

'Neural Architecture Search' workshop at ICLR 2021

Co-organizer

Jan 2021 – Dec 2021

'Neural Architecture Search' workshop at ICLR 2020

Co-organizer
Jan 2020 – Dec 2020

ICLR 2022

Highlighted Reviewer
Jan 2022 – Dec 2022

ICLR 2021

Outstanding Reviewer
Jan 2021 – Dec 2021

ICLR, ICML, NeurIPS, TMLR

Regularly Reviewer
Jan 2016 – Present

ELLIS

Member
Jan 2022 – Present

Education

University Bremen, Bremen, Germany

Computer Science
Nov 2009 – Feb 2014
Dr.rer.nat. (PhD)

University Münster, Münster, Germany

Computer Science
Oct 2001 – Aug 2006
Diploma

Projects

SVGStud.io

Jan 2024 – Present

As a hobby project, I have built SVGStud.io. SVGStud.io is a powerful AI SVG generator designed for creators, designers, and professionals looking to effortlessly generate, edit, and customize Scalable Vector Graphics (SVGs).

- SVGStud.io has more than 1000 weekly active users as of 03/2025.

Awards

Scholarship (2004 - 2006)

Awarded by **Studienstiftung des Deutschen Volkes** (German National Academic Foundation)
Jan 2004

Languages

German

Native speaker

Spanish

Basic

English

Fluent

French

Basic